

ACCT102 Management Accounting

Week 5

Absorption costing and Variable costing

Cost flows

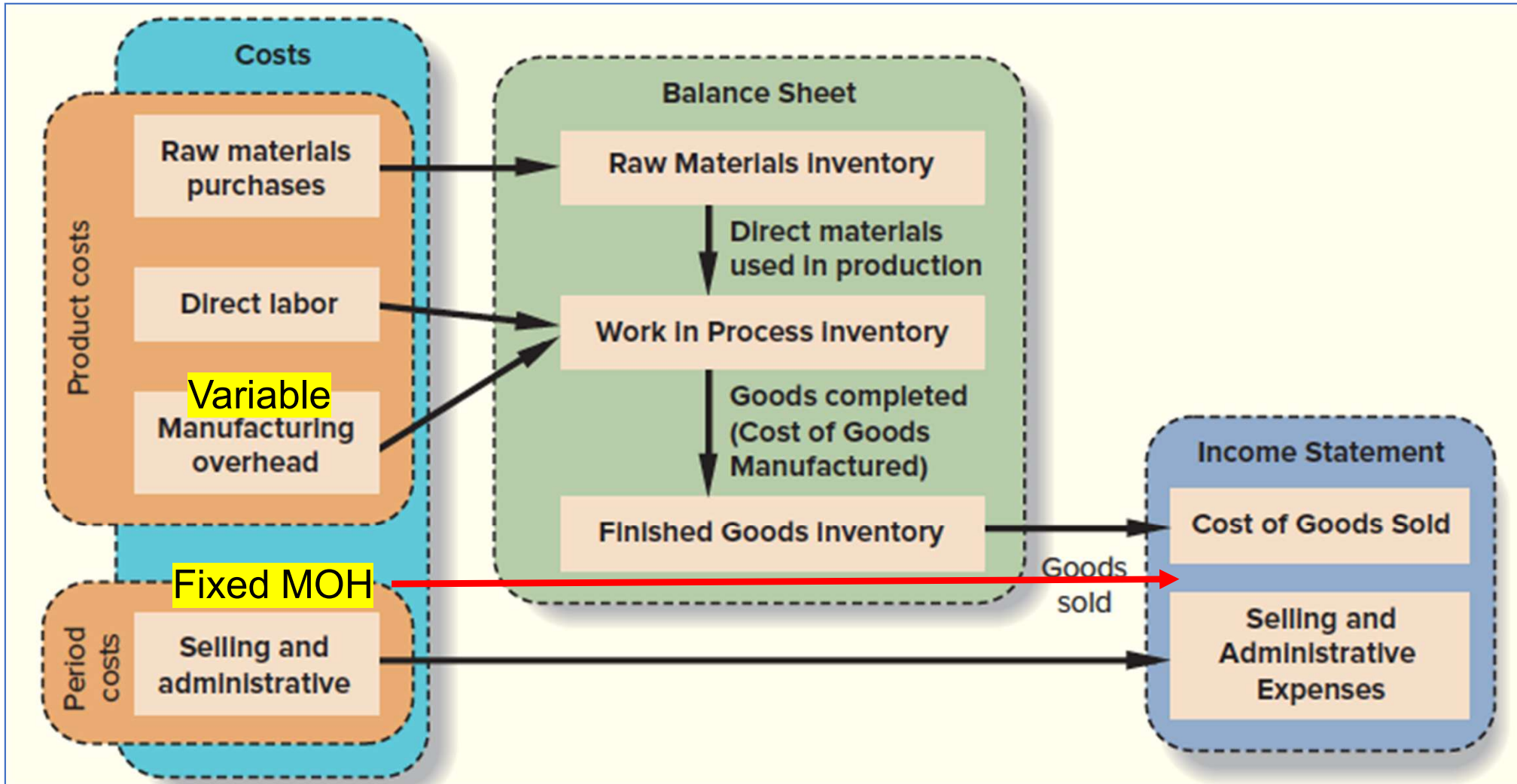
Variable and Absorption Costing Income Statements (Y1)

- In Year 1, Only 20,000 units were sold during the year, at a price of \$30 each.
- There is no beginning inventory.

Number of units produced annually	25,000
Variable costs per unit:	
Direct materials, direct labor, and variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3
Fixed costs per year:	
Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

<u>Unit product cost</u>	<u>Absorption Costing</u>	<u>Variable Costing</u>
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Fixed mfg. overhead	6	-
Unit product cost	\$ 16	\$ 10

VARIABLE COSTING



VARIABLE COSTING

COST

Product cost

\$10

10

x 25,000

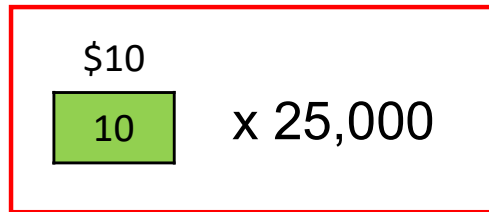
BALANCE SHEET

INCOME STATEMENT

VARIABLE COSTING

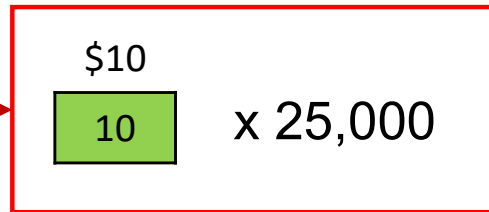
COST

Product cost



BALANCE SHEET

WIP inventory



INCOME STATEMENT

VARIABLE COSTING

COST

Product cost

$$\begin{array}{l} \$10 \\ \boxed{10} \end{array} \times 25,000$$

BALANCE SHEET

WIP inventory



FG inventory

$$\begin{array}{l} \$10 \\ \boxed{10} \end{array} \times 25,000$$

INCOME STATEMENT

VARIABLE COSTING

COST

Product cost

$$\begin{array}{c} \$10 \\ \boxed{10} \end{array} \times 25,000$$

BALANCE SHEET

WIP inventory

$$\boxed{}$$

FG inventory

$$\begin{array}{c} \$10 \\ \boxed{10} \end{array} \times 5,000$$

INCOME STATEMENT

COGS

$$\begin{array}{c} \$10 \\ \boxed{10} \end{array} \times 20,000$$



VARIABLE COSTING

COST

Product cost

\$10

10

x 25,000

BALANCE SHEET

WIP inventory

FG inventory

\$10

10

x 5,000

INCOME STATEMENT

COGS

\$10

10

x 20,000

Fixed MOH

\$150,000

S&A

\$3 / units sold

\$100,000

VARIABLE COSTING

COST

Product cost

$$\begin{array}{l} \$10 \\ \boxed{10} \end{array} \times 25,000$$

BALANCE SHEET

WIP inventory

$$\boxed{}$$

FG inventory

$$\begin{array}{l} \$10 \\ \boxed{10} \end{array} \times 5,000$$

INCOME STATEMENT

COGS

$$\begin{array}{l} \$10 \\ \boxed{10} \end{array} \times 20,000$$

Fixed MOH

$$\boxed{\$150,000}$$

Fixed MOH

$$\boxed{\$150,000}$$

S&A

$$\begin{array}{l} \$3 / \text{units sold} \\ \$100,000 \end{array}$$

S&A

$$\begin{array}{l} \$3 \times 20,000 \\ \$100,000 \end{array}$$

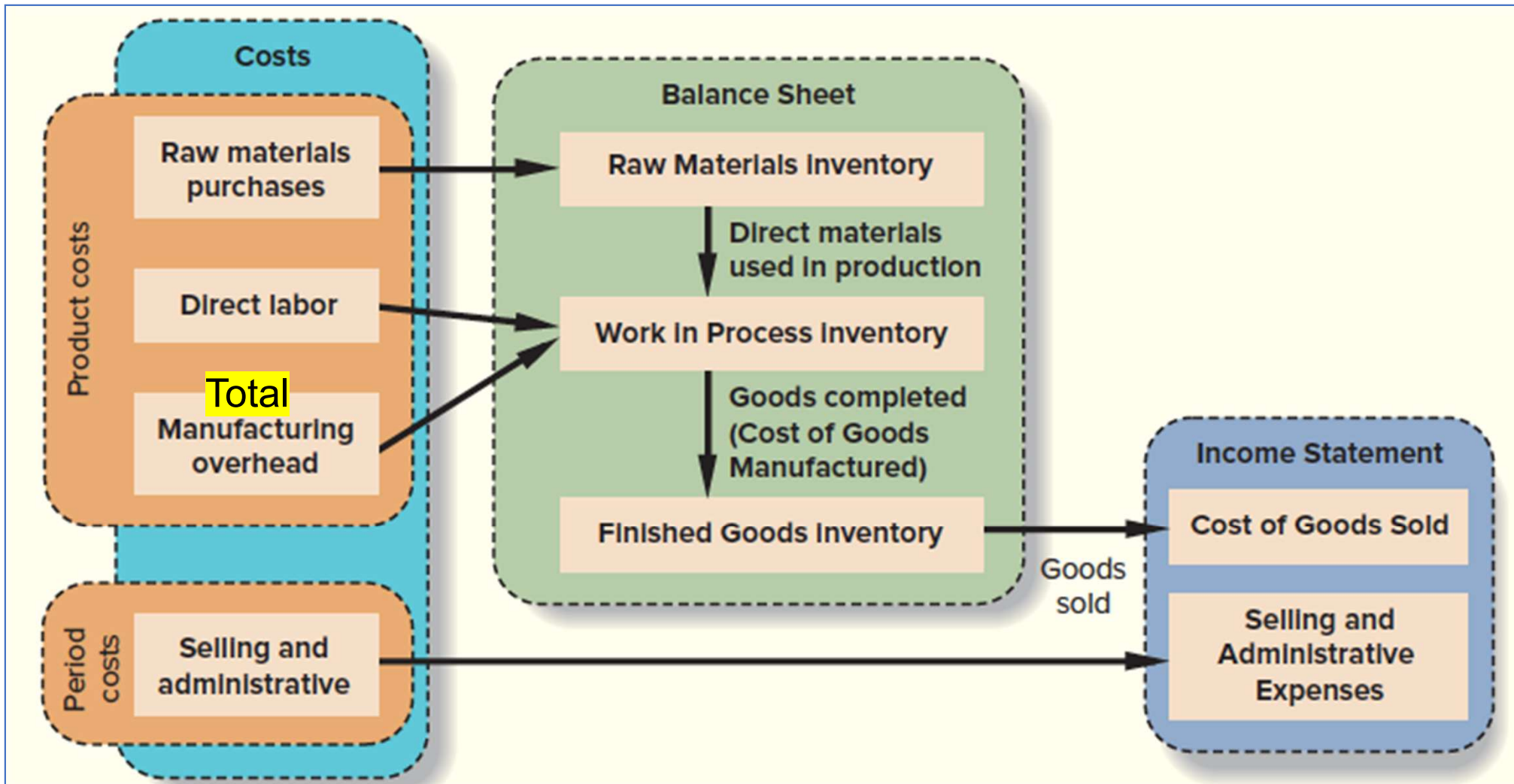
Variable Costing Income Statement (Y1)

	Variable Costing	
Sales (20,000 × \$30)		\$ 600,000
Less variable expenses:		
Variable cost of goods sold (20,000 × \$10)	\$ 200,000	
Variable selling & administrative expenses (20,000 × \$3)	60,000	
Total variable expenses		260,000
Contribution margin		340,000
Less fixed expenses:		
Fixed manufacturing overhead	\$ 150,000	
Fixed selling & administrative expenses	100,000	250,000
Net operating income		\$ 90,000

← Variable manufacturing costs only.

← Expense all fixed manu OH

Flow of Costs: Conceptual Overview



ABSORPTION COSTING

COST

Product cost

$$\begin{array}{l} \$10 \\ \boxed{10} \end{array} \times 25,000$$

$$\boxed{6} \times 25,000$$

Fixed MOH

$$\boxed{\$150,000}$$

BALANCE SHEET

INCOME STATEMENT

ABSORPTION COSTING

COST

Product cost

\$16	
10	x 25,000
6	

BALANCE SHEET

INCOME STATEMENT

ABSORPTION COSTING

COST

Product cost

\$16	
10	
6	
	x 25,000



BALANCE SHEET

WIP inventory

\$16	
10	
6	
	x 25,000

INCOME STATEMENT

ABSORPTION COSTING

COST

Product cost

\$16	
10	x 25,000
6	

BALANCE SHEET

WIP inventory

--

FG inventory

\$16	
10	x 25,000
6	

INCOME STATEMENT

ABSORPTION COSTING

COST

Product cost

\$16	
10	x 25,000
6	

BALANCE SHEET

WIP inventory

--

FG inventory

\$16	
10	x 5,000
6	

INCOME STATEMENT

COGS

\$16	
10	x 20,000
6	



ABSORPTION COSTING

COST

Product cost

\$16	
10	x 25,000
6	

BALANCE SHEET

WIP inventory

--

FG inventory

\$16	
10	x 5,000
6	

INCOME STATEMENT

COGS

\$16	
10	x 20,000
6	

S&A

\$3 / units sold
\$100,000

ABSORPTION COSTING

COST

Product cost

\$16	
10	x 25,000
6	

BALANCE SHEET

WIP inventory

--

FG inventory

\$16	
10	x 5,000
6	

INCOME STATEMENT

COGS

\$16	
10	x 20,000
6	

S&A

\$3 / units sold
\$100,000

S&A

\$3 x 20,000
\$100,000



Absorption Costing Income Statement (Y1)

Absorption Costing			
Sales (20,000 × \$30)		\$	600,000
Cost of goods sold (20,000 × \$16)			320,000
Gross margin			280,000
Less selling & administrative expenses:			
Variable (20,000 × \$3)	\$	60,000	
Fixed		100,000	160,000
Net operating income		\$	120,000

← Includes fixed manu OH

The rest of the fixed manufacturing overhead is deferred in inventory.
 5,000 units (produced but not sold) × \$6 = \$30,000.

Variable and Absorption Costing Income Statements (Y2)

- In Year 2, 30,000 units were sold during the year, at a price of \$30 each.
- There is a beginning inventory of 5,000 units from Year 1.

	YEAR 1	YEAR 2
Number of units produced	25,000	25,000
Number of units sold	20,000	30,000
Units in beginning inventory	-	5,000
Unit sales price	\$ 30	\$ 30
Variable costs per unit:		
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Selling & administrative expenses	\$ 3	\$ 3
Fixed costs per year:		
Manufacturing overhead	\$ 150,000	\$ 150,000
Selling & administrative expenses	\$ 100,000	\$ 100,000

	AC	VC
Unit product cost	\$16	\$10

VARIABLE COSTING

COST

Product cost

$$\begin{array}{c} \$10 \\ \boxed{10} \end{array} \times 25,000$$

BALANCE SHEET

WIP inventory

$$\begin{array}{c} \$10 \\ \boxed{10} \end{array} \times 0$$

FG inventory

$$\begin{array}{c} \$10 \\ \boxed{10} \end{array} \times 0 \quad \text{Initial: 5,000}$$

INCOME STATEMENT

COGS

$$\begin{array}{c} \$10 \\ \boxed{10} \end{array} \times 30,000$$

Fixed MOH

$$\$150,000$$

Fixed MOH

$$\$150,000$$

S&A

$$\begin{array}{c} \$3 / \text{units sold} \\ \$100,000 \end{array}$$

S&A

$$\begin{array}{c} \$3 \times 30,000 \\ \$100,000 \end{array}$$

Variable Costing Income Statement (Y2)

	Variable Costing	
Sales (30,000 × \$30)		\$ 900,000
Less variable expenses:		
Variable cost of goods sold (30,000 × \$10)	\$ 300,000	← <i>Variable manufacturing costs only.</i>
Variable selling & administrative expenses (30,000 × \$3)	90,000	
Total variable expenses		390,000
Contribution margin		510,000
Less fixed expenses:		
Fixed manufacturing overhead	\$ 150,000	← <i>Expense all fixed manu OH</i>
Fixed selling & administrative expenses	100,000	250,000
Net operating income		260,000

ABSORPTION COSTING

COST

Product cost

\$16	
10	x 25,000
6	

BALANCE SHEET

WIP inventory

--

FG inventory

\$16	Initial: 5,000
10	x 0
6	

INCOME STATEMENT

COGS

\$16	
10	x 30,000
6	

S&A

\$3 / units sold
\$100,000

S&A

\$3 x 30,000
\$100,000

Absorption Costing Income Statement (Y2)

Absorption Costing			
Sales (30,000 × \$30)		\$	900,000
Cost of goods sold (30,000 × \$16)	← <i>Includes fixed manu OH</i>		480,000
Gross margin			420,000
Less selling & administrative expenses:			
Variable (30,000 × \$3)		\$	90,000
Fixed		100,000	190,000
Net operating income		\$	230,000

Fixed manufacturing overhead is released from beginning inventory balance 5,000 units (sold in Y2 but produced in Y1) × \$6 = \$30,000.